

```
In[*]:= v = ((299 792 458) / ({34.4, 37.1, 40, 43.7, 48.1} * 10^(-12))) /
(10^(18)) (*EHz = 10^18 Hz*)
```

```
Out[*]:=
{8.7149, 8.08066,  $\frac{149\,896\,229}{20\,000\,000}$ , 6.86024, 6.23269}
```

```
In[*]:= V = {35, 32.5, 30, 27.5, 25}; (*kV*)
```

```
In[*]:= data = Transpose[{v, V}]
└transposición
```

```
Out[*]:=
{{8.7149, 35}, {8.08066, 32.5}, { $\frac{149\,896\,229}{20\,000\,000}$ , 30}, {6.86024, 27.5}, {6.23269, 25}}
```

```
In[*]:= fit = NonlinearModelFit[data, A * x + B, {A, B}, x]
└ajusta a modelo no lineal
```

```
Out[*]:=
FittedModel[ $-0.217 + 4.04 x$ ]
```

```
In[*]:= fit[{"BestFit", "ParameterTable"}]
```

```
Out[*]:=
{ $-0.217068 + 4.04152 x$ , 

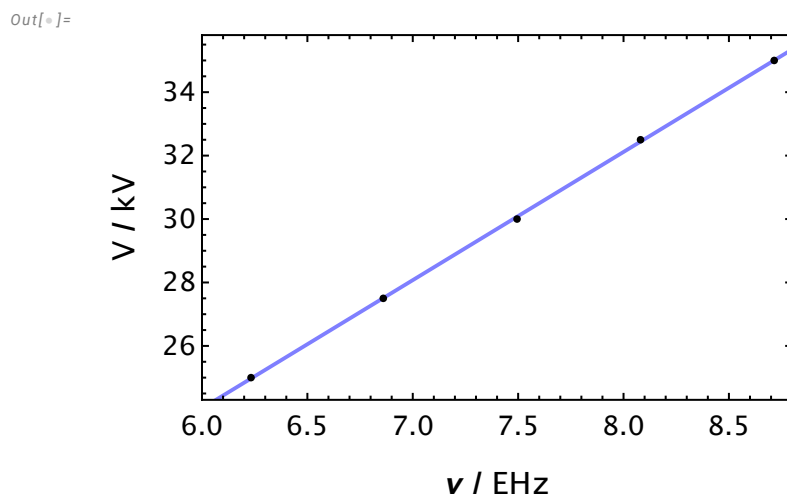
|   | Estimate  | Standard Error | t-Statistic | P-Value                  |
|---|-----------|----------------|-------------|--------------------------|
| A | 4.04152   | 0.0290809      | 138.975     | $8.21445 \times 10^{-7}$ |
| B | -0.217068 | 0.218911       | -0.991582   | 0.394498                 |

}
```

```
In[*]:= plotfit =
Plot[fit[x], {x, 0, 100}, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
└representación gráfica └estilo base └familia de tipo de letra └tamaño de tipo de letra
PlotStyle -> {Blend[{Blue, White}], Thick}, PlotRange -> {{6.0, 8.8}, {24.3, 35.8}},
└estilo de repre... └mezcla... └azul └blanco └grueso └rango de representación
Frame -> True, FrameLabel -> {Style["v / EHz", Bold, 16], Style["V / kV", 16, Bold]};
└marco └verd... └etiqueta de marco └estilo └negrita └estilo └negrita
```

```
In[*]:= plotdata = ListPlot[data, PlotStyle -> {Black, Thickness[0.003]};
└representación de... └estilo de repre... └negro └grosor
```

```
In[*]:= Show[plotfit, plotdata]
└muestra
```



```
In[*]:= Aaj = 4.041519791400553;
```

```
In[*]:= errAj = 0.029080879127361545;
```

```
In[*]:= h = Aaj * (1.602 * 10^(-19)) * (10^3) / (10^18) (*J s*)
```

```
Out[*]=  
 $6.47451 \times 10^{-34}$ 
```

```
In[*]:= errh = errAj * (1.602 * 10^(-19)) * (10^3) / (10^18) (*J s*)
```

```
Out[*]=  
 $4.65876 \times 10^{-36}$ 
```